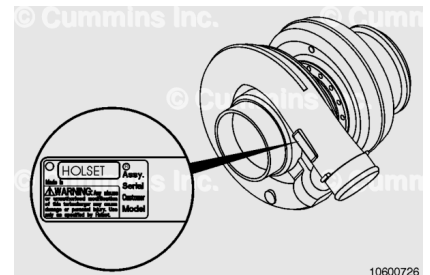


Trainings ▾

General Information

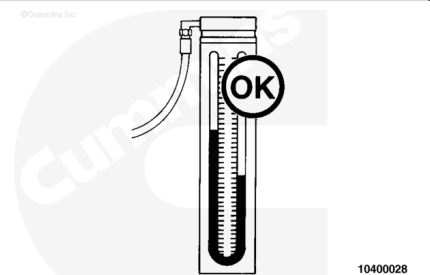
The turbochargers used on the QST30 engines can differ in the service methods based on their applications. Locate the turbocharger dataplate to verify the procedures contained within this procedure apply and the correct specifications are used.



Initial Check

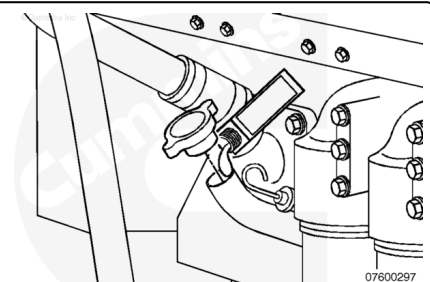
Inspect the engine and turbocharger area for any external leaks.

Measure the crankcase pressure (blowby). Refer to Procedure 014-005 in Section 14.
(</qs3/pubsys2/xml/en/procedures/57/57-014-005.html>)



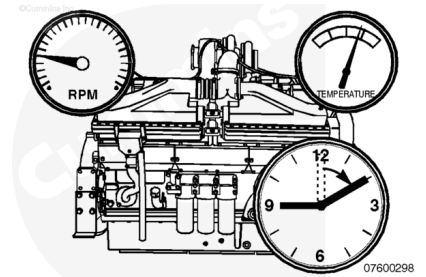
Fluorescent Dye Tracer

Add contents of one package (4 ounces) of fluorescent tracer, Part Number 3376891, or equivalent, to each 38 liters [10 U.S. gal] of engine lubricating oil.

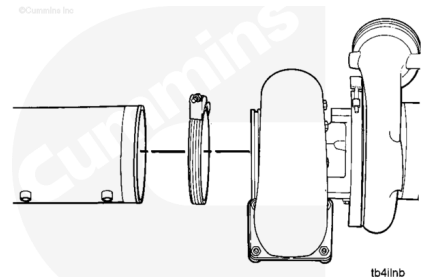


Operate the engine at low idle for 10 minutes.

Shut the engine OFF.

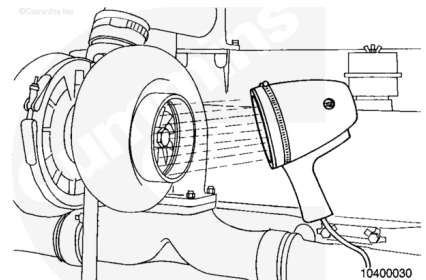


Allow the turbocharger to cool and remove the exhaust pipe from the turbine housing.



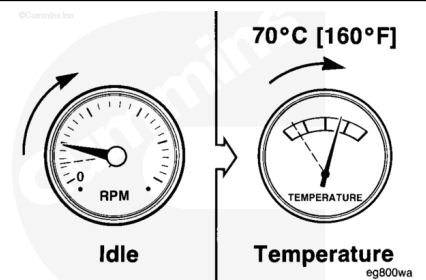
Use a high-intensity black light to inspect the turbine housing outlet for oil.

A dark blue glow indicates fuel slobber and a yellow glow indicates oil carryover.



The coolant temperature needs to be maintained at a minimum of 71°C [160°F].

If fuel is found in the turbine housing, the problem is excess, unburned fuel. To correct the problem, decrease the idle period, or increase the idle rpm.



If oil is found in the turbine housing, disconnect the oil drain tube from the cylinder block.

Remove the oil drain and supply lines from the turbocharger.

Refer to Procedure 010-045 in Section 10.

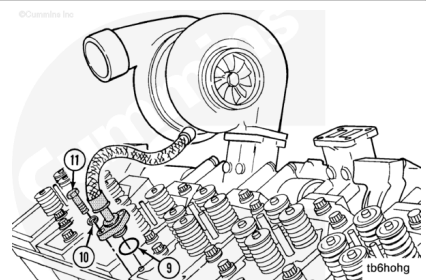
(/qs3/pubsys2/xml/en/procedures/57/57-010-045.html) Refer to Procedure 010-046 in Section 10.

(/qs3/pubsys2/xml/en/procedures/57/57-010-046.html)

Remove any support clamps and the hose.

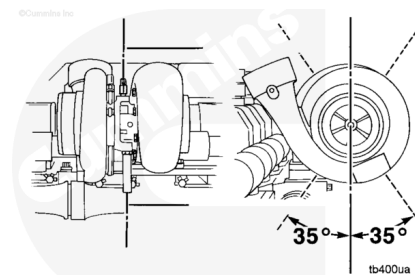
Remove and discard the o-ring gasket.

Clear any restrictions found in the hose.

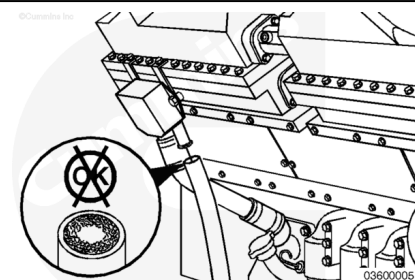


Discard the hose if it is cracked, frayed, or will **not** bend without cracking.

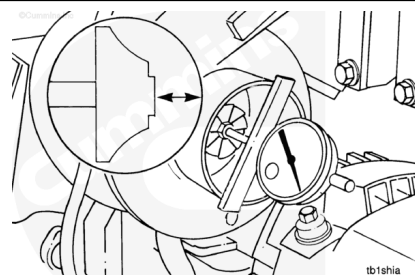
Check the angle of the drain tube. The angle of the tube **must** be within 35 degrees of vertical. Adjust the turbocharger, if necessary. Reference the Bearing Housing Adjustment in the Install section in this procedure.



If the drain tube is free of restrictions and at the correct angle, check the crankcase breather and tube to verify they are **not** plugged or restricted.

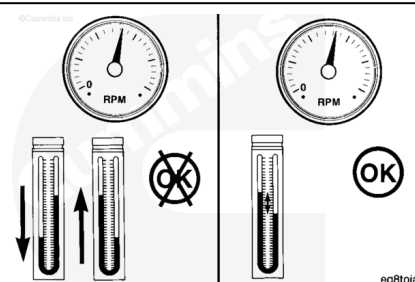


Check the turbocharger axial clearance and radial clearance. Reference to Measure Compressor Impeller Endplay in the Inspect for Reuse section of this procedure.



Install the exhaust pipe into the turbine housing.

If these checks do **not** reveal the problem, measure the crankcase (blowby). Refer to Procedure 014-005 in Section 14. (</qs3/pubsys2/xml/en/procedures/57/57-014-005.html>)



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

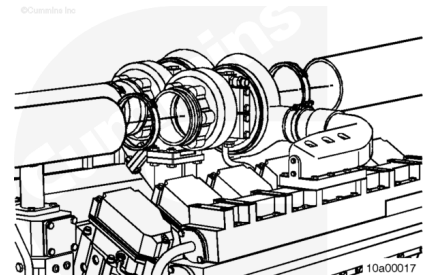
⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the turbocharger oil supply line. Refer to Procedure 010-046 in Section 10. (</qs3/pubsys2/xml/en/procedures/57/57-010-046.html>)
- Remove the turbocharger oil drain line. Refer to Procedure 010-045 in Section 10. (</qs3/pubsys2/xml/en/procedures/57/57-010-045.html>)

Remove

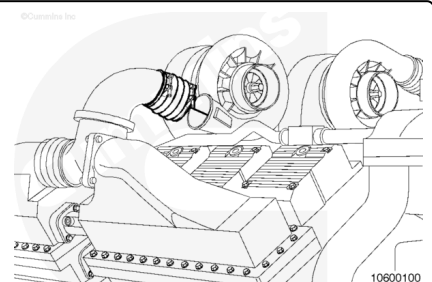
Remove the intake and exhaust pipes from the turbocharger.



Remove the hose and two clamps between the air crossover and the turbocharger.

Remove the clamp between the air crossover and the aftercoolers or intercoolers.

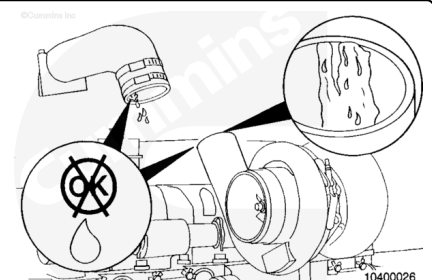
Remove the air crossover.



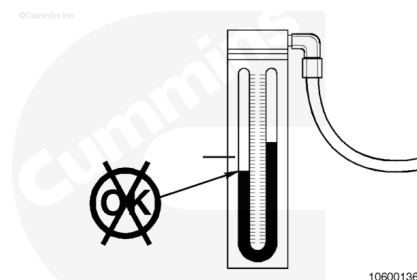
Inspect the compressor discharge and air crossover piping for oil.

If oil is in the compressor discharge or crossover piping, check the turbocharger axial clearance and radial clearance.

Reference to Measure Compressor Impeller Endplay in the Inspect for Reuse section of this procedure.

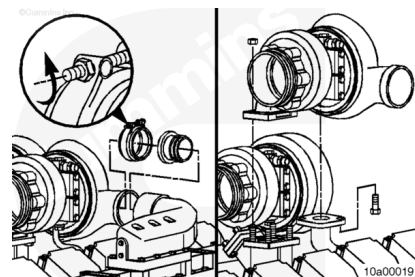


Check for intake restriction. Refer to Procedure 010-031 in Section 10. (/qs3/pubsys2/xml/en/procedures/57/57-010-031.html)



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove the V-band clamp, discharge elbow, and o-ring from the turbocharger compressor discharge outlet.

Remove the four turbocharger mounting nuts.

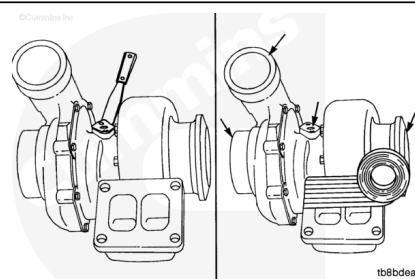
Remove the turbocharger and gasket.

Discard the gasket.

Clean

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Tape or plug all openings to prevent solvent or steam from entering the oil ports in the turbocharger.

Clean the exterior of the turbocharger and exhaust manifold gasket surfaces with solvent or steam.

Steam clean the turbocharger.

Dry with compressed air.

Inspect for Reuse

Remove the tape or plugs from the turbocharger openings.

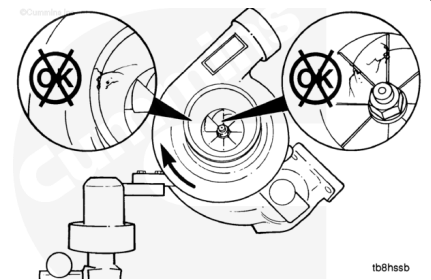
Inspect the turbine and compressor wheels for fretting.

Check the vanes for cracks or other damage.

Repair or replace the turbocharger if the vanes are damaged.

Inspect the exterior for any damage.

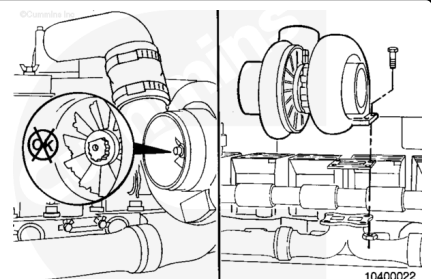
Note : If inspections or dimensional checks indicate a problem, contact a Cummins® Authorized Repair Location for assistance. Reference the model number on the turbocharger dataplate.



Inspect the turbocharger compressor impeller blades for damage.

Rebuild or replace the turbocharger if damage is found.

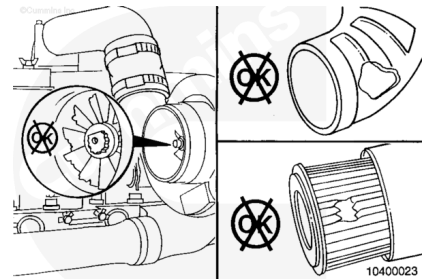
Contact a Cummins® Authorized Repair Location.



Note : Repair any damage before operating the engine.

If the impeller is damaged, inspect the intake air piping and the filter element for damage.

Inspect the compressor cover and intake manifold for any debris.



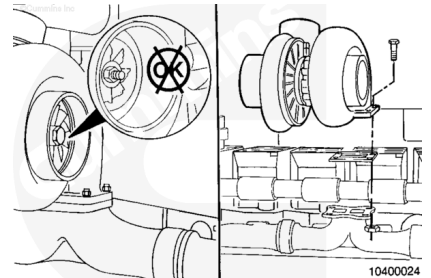
Inspect the turbine wheel for damage.

Check for debris stuck in the turbine housing.

If damage or debris is present, check for debris in the exhaust manifold.

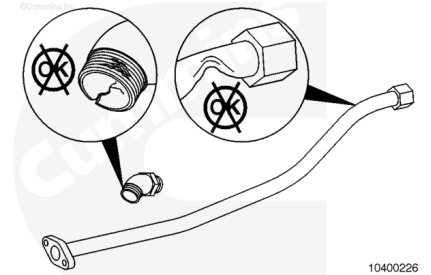
Rebuild or replace the turbocharger if damage is found.

Contact a Cummins® Authorized Repair Location.



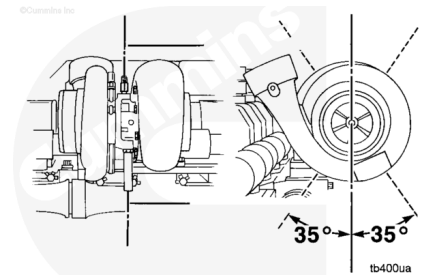
If oil is found in the turbine housing, inspect the oil drain tube and clear any restrictions found in the hose.

Discard the hose if it is cracked, frayed, or will **not** bend without cracking.

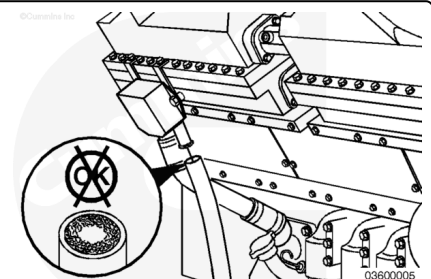


Check the angle of the drain tube.

The angle of the tube **must** be within 35 degrees of vertical. Adjust the turbocharger, if necessary. Reference Bearing Housing Adjustment in the Install section of this procedure.

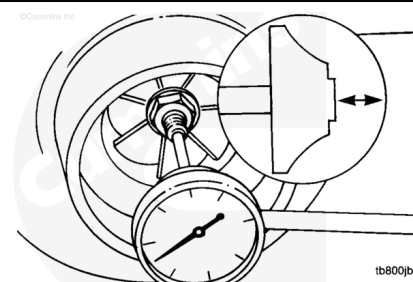


If the drain tube is free of restrictions and at the correct angle, check the crankcase breathers and tubes to make sure they are **not** plugged or restricted.



Measure

Use a depth gauge, Part Number ST-537, or a dial indicator to measure the axial motion (end-to-end).



Compressor Impeller Axial End Clearance			
	mm		in
Holset® HX82/HX8 3 Turbocharg er	0.025	MIN	0.001
	0.152	MAX	0.006
Holset® HX60 Turbocharg er	0.051	MIN	0.002
	0.152	MAX	0.006
Holset® HE851/HE 800 Turbocharg er	0.025	MIN	0.001
	0.152	MAX	0.006

Make sure the movement is within the MIN/MAX Total Indicator Reading (TIR) values shown above.

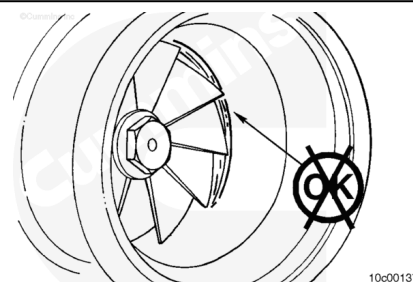
Note : If the end clearance exceeds the specifications, the turbocharger must be replaced or rebuilt. Contact a Cummins® Authorized Repair Location.

Check the radial movement of the turbocharger wheels and shaft.

Use light finger pressure to push the compressor wheel toward the side of the compressor housing.

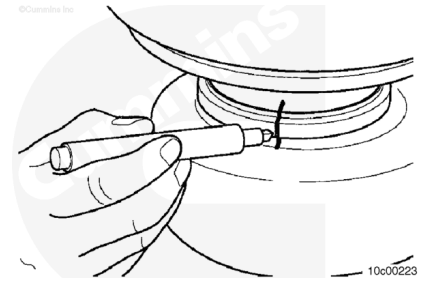
Repeat the procedure for the turbine wheel and housing.

Replace the turbocharger if either wheel contacts the housing.

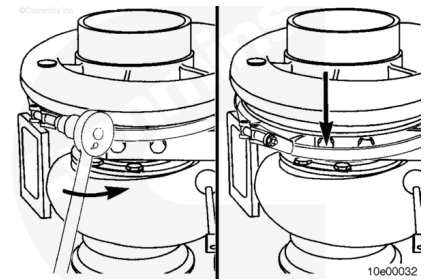


Disassemble

Place the turbine outlet on a clean flat surface. Mark the turbine housing, center housing, rotating assembly, V-band clamp, and compressor housing. This action assists in the reassembly process defining correct component orientation.

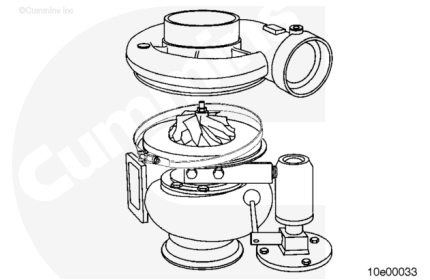


Remove V-band clamp(s). Discard V-band clamps and locknut(s).



⚠ CAUTION ⚠

Use care when handling the turbocharger when the compressor housing is removed. The compressor impeller blades can be easily damaged.

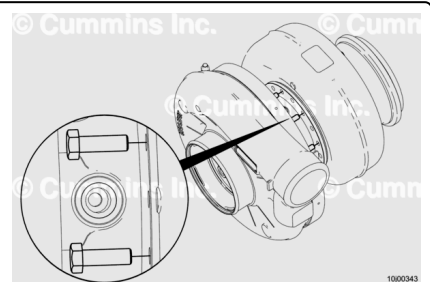


Remove the compressor housing.

Use a hammer and a chisel to bend the lockplate tabs, if equipped, away from the turbine housing capscrews.

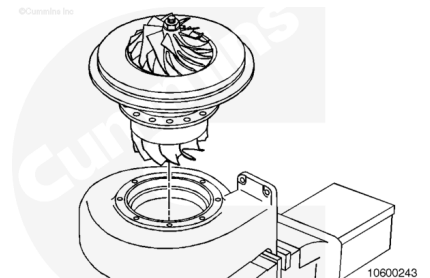
Remove and discard the turbine housing capscrews lockplates, if equipped.

Remove and discard the turbine housing capscrews, if equipped.



⚠ CAUTION ⚠

Use care when separating the center housing rotating assembly from the turbine housing. The turbine blades can be easily damaged when the center housing rotating assembly is separated from the turbine housing.

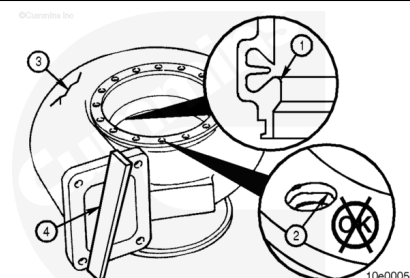


Separate the center housing rotating assembly from the turbine housing.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Do not bead blast an area of a component for a long period of time. Bead blasting areas for a long period of time can affect the component balance.

Inspect the turbine housing for signs of turbine impeller rub (1).

If any signs of turbine impeller rubbing are found, the turbine housing **must** be replaced.

Inspect the tapped holes for damaged threads (2).

Repair damaged threads or replace the turbine housing.

Check for signs of cracking or flaking caused by possible overheating (3).

Replace the turbine housing if it is cracked.

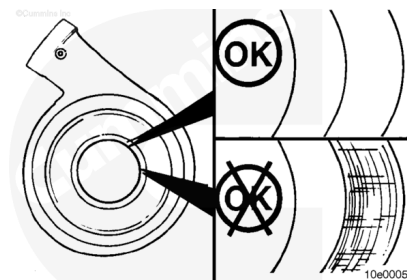
Check the flange for flatness.

Turbocharger Turbine Housing Flange Flatness		
mm		in
0.102	MIN	0.004

If the turbocharger turbine housing flange is **not** within specifications, the turbocharger turbine housing **must** be repaired or replaced.

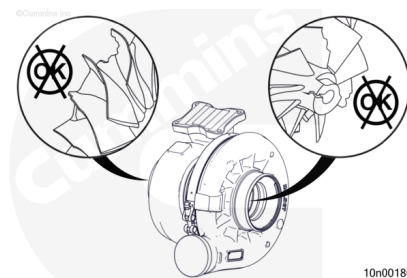
Check the compressor housing for signs of the compressor impeller rubbing.

If any signs of the compressor impeller rubbing are found, replace the compressor housing.



Inspect the compressor impeller for bent or cracked blades, or blades worn at the end.

Replace the turbocharger if any damage or wear is found.



Assemble

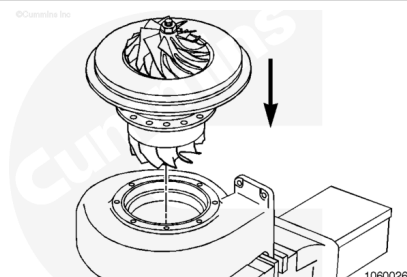
⚠ CAUTION ⚠

Use care when installing the center housing rotating assembly into the turbine housing. The turbine wheel blades can be easily damaged.

Set the turbine housing on a clean flat surface.

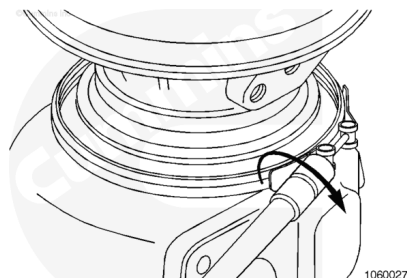
Place a new v-band clamp into position loosely on the turbine housing, if equipped.

Carefully slide the center housing rotating assembly into the turbine housing.



If reinstalling turbocharger in same location and alignment marks are available:

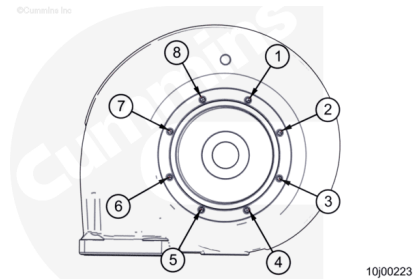
- Align the marks on the turbine housing to the center housing rotating assembly.
- Position the turbine V-band clamp, if equipped, on the center housing rotating assembly and turbine housing.
- Install a new locknut and tighten.



Torque Value:

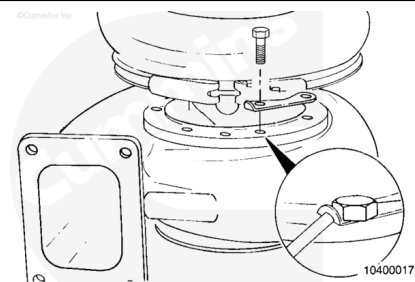
Holset® HX55 and HX60 9 n•m [80 in-lb]

- Install lockplate tabs, if equipped.
- Apply anti-seize compound, Part Number 3824879, to the capscrews, if equipped.
- Install turbocharger turbine housing mounting capscrews.



Turbine housing bolts **must** be double tightened. Tighten bolts once in the circular sequence shown and repeat tightening each bolt again to the same value in the same circular sequence.

HX82	Torque Value:	Step 1	22 n•m [195 in-lb]
		Step 2	22 n•m [195 in-lb]
HX83	Torque Value:	Step 1	20 n•m [177 in-lb]
		Step 2	20 n•m [177 in-lb]
HE851 (with lockplates)	Torque Value:	Step 1	20 n•m [177 in-lb]
		Step 2	20 n•m [177 in-lb]
HE800/HE851 (no lockplates)	Torque Value:	Step 1	26 n•m [230 in-lb]
		Step 2	26 n•m [230 in-lb]



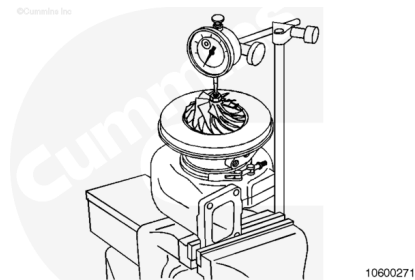
Bend the lockplate tabs over the capscrews, if equipped.

If installing turbocharger in a different location or alignment marks are **not** available:

- Install turbocharger turbine housing mounting capscrews.
- Do **not** tighten capscrews until proper orientation is confirmed during installation of turbocharger.

Make sure the turbine shaft rotates freely.

Check the turbocharger shaft end clearance.



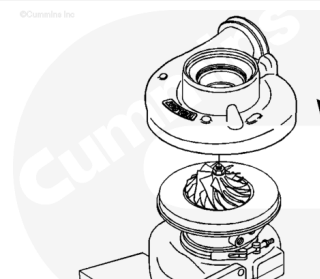
Holset® HX60 Turbocharger Shaft End Clearance

	mm		in
--	----	--	----

Compress or Impeller	0.051	MIN	0.002
	0.152	MAX	0.006

⚠ CAUTION ⚠

Use care when installing the compressor housing onto the turbocharger. The compressor impeller can be easily damaged.



10600273

Place the turbocharger turbine housing on a flat clean surface.

Place the compressor housing V-band clamp loosely onto the center housing rotating assembly.

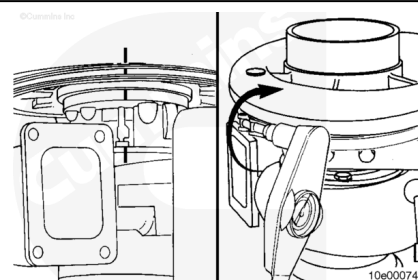
Carefully place the compressor housing over the compressor impeller onto the center housing rotating assembly.

Rotate compressor housing to proper orientation for location turbocharger will be installed.

Install new compressor housing V-band clamp.

If reinstalling turbocharger in same location and alignment marks are available:

- Tighten V-band clamp



10600074

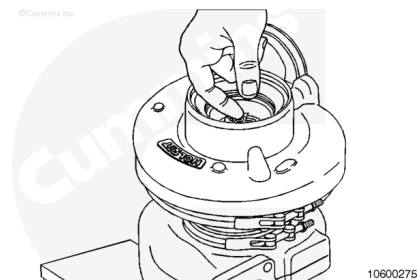
HX55/HX60	Torque Value	9 n•m [80 in-lb]
HX82	Torque Value	12 n•m [106 in-lb]
HX83	Torque Value	12 n•m [106 in-lb]
HE851 (with lockplates)	Torque Value	12 n•m [106 in-lb]
HE800/HE851 (no lockplates)	Torque Value	12 n•m [106 in-lb]

If installing turbocharger in a different location or alignment marks are **not** available:

- Tighten V-band clamp enough compressor housing will **not** rotate during installation.

Make sure the turbine shaft rotates freely.

Note : Some turbochargers have a coating applied to the internal surface of the turbine housing. This special coating is abraded by the turbine impeller and can affect turbocharger rotation until its unique form is fully formed.



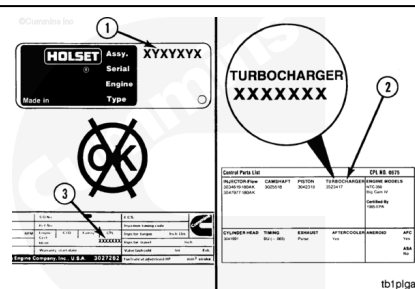
Install

Check for Correct Component

Compare the assembly number (1) on the turbocharger dataplate with the turbocharger specified in the Control Parts List, Bulletin 3379133 (2).

The CPL number for each engine is listed on the engine's dataplate (3).

Reference the data tag on the turbocharger to determine the model.

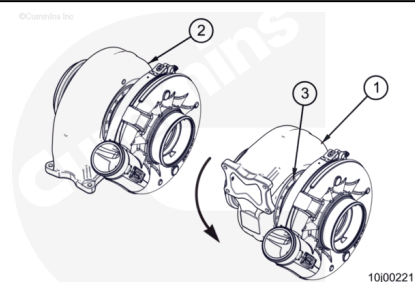


New turbocharger pre-installation check:

Check the exterior of the turbocharger for possible shipping damage or excessive corrosion. Flash corrosion can be cleaned off with a towel.

Inspect for corrosion on the turbocharger impeller wheel and on visible surfaces within the compressor and turbine housing.

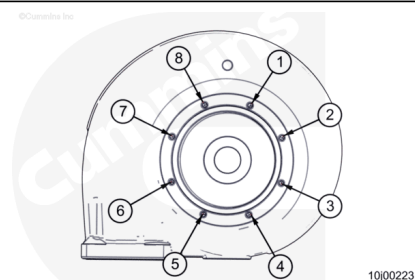
If installing new turbocharger (1), compare orientation of turbine housing and compressor housing to the turbocharger being replaced (2).



If turbine housing orientation with respect to bearing housing does **not** match turbocharger being replaced, place turbocharger on turbine housing outlet flange on clean sturdy work bench.

Secure the turbocharger and remove turbine housing to bearing housing mounting capscrews (3).

Rotate turbine housing to match orientation of turbocharger being replaced.



Install turbine housing to bearing housing V-band clamp, if equipped.

Torque Value:

Holset® HX55 and HX60 9 n•m [80 in-lb]

Install lockplate tabs, if equipped.

Apply anti-seize compound, Part Number 3824879, to the capscrews.

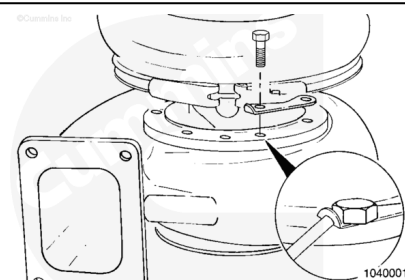
Install turbine housing to bearing housing mounting capscrews, if equipped.

Tighten capscrews in sequence shown.

Turbine housing bolts shall be double tightened. Tighten bolts once in the circular sequence shown and repeat tightening each bolt again to the same value in the same circular sequence.

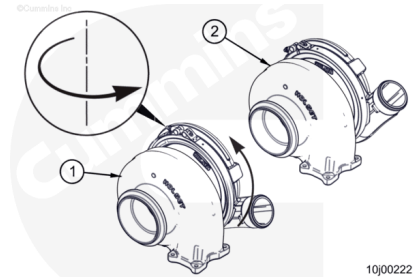
HX82	Torque Value:	Step 1	22 n•m [195 in-lb]
		Step 2	22 n•m [195 in-lb]
HX83	Torque Value:	Step 1	20 n•m [177 in-lb]
		Step 2	20 n•m [177 in-lb]
HE851 (with lockplates)	Torque Value:	Step 1	20 n•m [177 in-lb]
		Step 2	20 n•m [177 in-lb]
HE800/HE851 (no lockplates)	Torque Value:	Step 1	26 n•m [230 in-lb]
		Step 2	26 n•m [230 in-lb]

Bend the lockplate tabs over the capscrews, if equipped.



⚠ CAUTION ⚠

Axial movement of compressor housing can result in component damage.



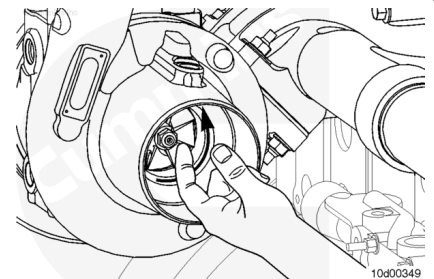
On new turbocharger (1), if compressor housing orientation with respect to bearing housing does **not** match turbocharger being replaced (2), loosen V-band clamp between turbocharger bearing housing and turbocharger compressor housing.

Rotate compressor to match orientation of turbocharger being replaced.

Tighten V-band clamp enough compressor housing will **not** rotate during installation.

⚠ CAUTION ⚠

Remove the plastic shipping plugs in turbocharger bearing housing for oil supply and oil drain or turbocharger damage may result.



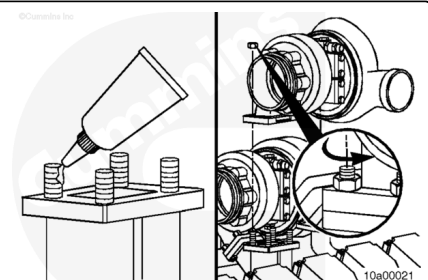
If installing new turbocharger, remove plastic shipping plugs from turbocharger bearing housing.

Lubricate the bearings by pouring 59 to 89 ml [2 to 3 fl-oz] of clean 15W-40 engine oil into the turbocharger oil supply line fitting. Rotate the turbine wheel to allow the oil to enter the bearing housing.

Note : If too much oil is poured into the bearing housing, the oil may leak past the seals. This is **not** regarded as a malfunction, since the seals are not dynamic and rely on exhaust gas and boost pressure to provide a proper seal.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Inspect the engine exhaust, intake, and aftercooler systems are clean and without obstruction.

Note : Orientation of the oil supply lines designates right or left bank turbocharger.

Apply a film of high-temperature anti-seize compound to the turbocharger mounting studs.

Use a new gasket and install the turbocharger.

Note : The word **OUT** on the turbocharger gasket is installed toward the turbocharger.

Note : Check that the exhaust manifold flange is flat and undamaged. When mounting the turbocharger on the flange, check that the turbine inlet gasket fits properly without obstructing the gas passages.

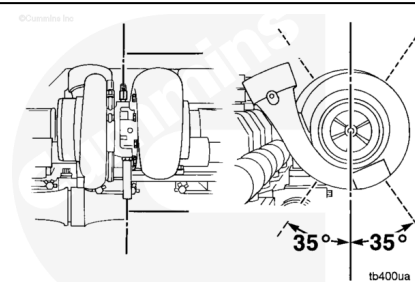
Install and tighten the four mounting nuts.

Note : The machined surface of the nut is installed against the turbocharger flange.

Note : The torque values given have been established using anti-seize compound as a lubricant.

Torque Value: 57 n•m [42 ft-lb]

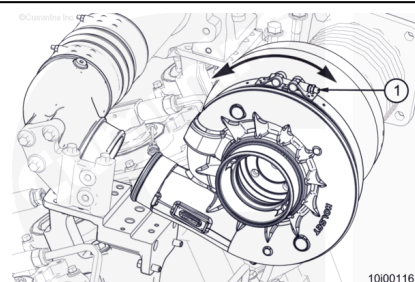
Position the turbocharger drain tube. The drain tube **must** be within 35 degrees of vertical. Turn the bearing housing to align the tube, if necessary.



Rotate turbocharger compressor housing by loosening V-band (1) between turbocharger bearing housing and turbocharger compressor housing.

Rotate turbocharger compressor housing to align with air crossover connection.

Tighten V-band clamp.

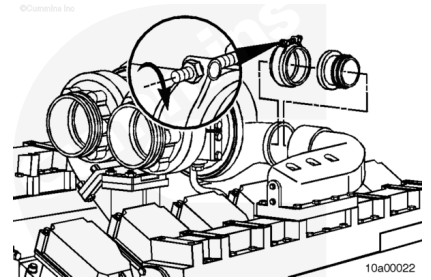


HX55/HX60	Torque Value	9 n•m [80 in-lb]
HX82	Torque Value	12 n•m [106 in-lb]
HX83	Torque Value	12 n•m [106 in-lb]
HE851 (with lockplates)	Torque Value	12 n•m [106 in-lb]
HE800/HE851 (no lockplates)	Torque Value	12 n•m [106 in-lb]

Install the discharge elbow, V-band clamp, and new o-ring seal on the turbocharger compressor discharge outlet.

Tighten the clamp.

Torque Value: 8 n•m [71 in-lb]



Note : The turbocharger end of the drain line has a gasket. The block end of the drain line has an o-ring.

Note : Mount the crook in oil drain tube toward the compressor housing.

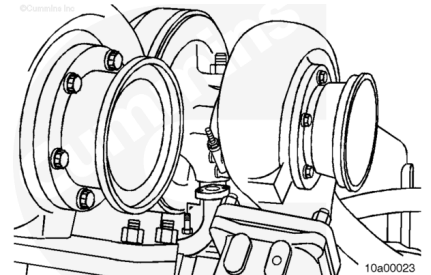
Press the o-ring end of the drain tube into the block.

Lubricate the o-ring with vegetable oil.

Note : The capscrew used for the right-bank turbocharger oil supply tube to block mounting is also used for the right-bank high-pressure tube clamp.

Tighten the capscrews.

Torque Value: 65 n•m [48 ft-lb]



Note : Check that the turbocharger oil inlet and drain pipes and connectors are clean, free from obstruction, and will not leak under pressure. Before re-installing flexible pipes, always make sure that any burned on lacquer or other adhered material is removed from the internal bores. If in doubt, fit new pipes.

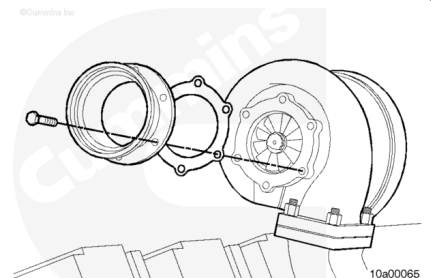
Note : Check that the coolant pipes of water cooled bearing housing applications and connectors are clean, free from obstruction, and will not leak under pressure. If in doubt, fit new pipes.

Install the gasket and exhaust outlet adapter to the turbocharger.

Apply anti-seize compound to the capscrews.

Tighten the capscrews.

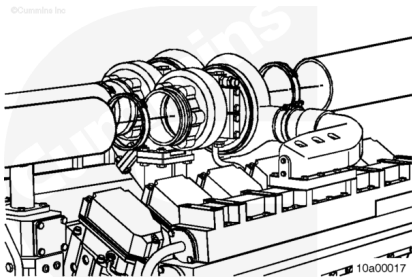
Torque Value: 50 n•m [37 ft-lb]



Install the intake pipe and tighten the clamp.

Install the exhaust pipe and tighten the clamp.

Torque Value: 8 n•m [71 in-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the turbocharger oil drain line. Refer to Procedure 010-045 in Section 10. (</qs3/pubsys2/xml/en/procedures/57/57-010-045.html>)
- Install the turbocharger oil supply line. Refer to Procedure 010-046 in Section 10. (</qs3/pubsys2/xml/en/procedures/57/57-010-046.html>)
- Connect the batteries. See equipment manufacturer service information.
- For applicable applications, start the engine and run at idle speed for approximately one minute so that the oil supply system is fully operational.
- Operate the engine at 71°C [160°F] coolant temperature and check for leaks.
- Accelerate the engine and check that there are no leaks/ obstructions of air, oil, coolant, or gas under pressure.